Problem-Solution Fit

Date	30 September 2026
Team ID	SWTID-2026-2032
Project Name	Learnmate – AI StudyBuddy
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

This canvas checks that Learnmate accurately addresses the root causes, behaviors, and limitations of its target audience.

1. CUSTOMER SEGMENT(S) [CS]

- **Students** (college and self-learners) preparing for exams with long PDFs, lecture notes and text material
- **Admins** who manage users and uploaded materials on the platform

6. CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES [CL]

- Very little time before exams
- Little or no budget for paid tutors or tools
- Phone or basic laptop, limited internet
- Files must be .txt, .md or .pdf up to 5 MB; scanned or image-only files not supported yet

5. AVAILABLE SOLUTIONS
PROS & CONS [AS]

- **Manual notes:** free and personal, but slow
- **Generic AI chatbots:** quick, but results are not saved and have no flashcards, quizzes or plan
- **Flashcard apps (e.g., Anki, Quizlet):** good for revision, but cards are made by hand

2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

- Long material is hard to condense into key points • Making flashcards and quizzes by hand takes hours
- No clear day-by-day study plan
- **Frequency:** daily during exam periods, weekly otherwise

9. PROBLEM ROOT / CAUSE [RC]

- Too much content and too little time
- No easy tool that turns a student's own material into summary, practice questions and a plan
- Students are unsure which topics matter most

7. BEHAVIOR
+ ITS INTENSITY [BE]

- Re-reads the same notes many times
- Crams close to the exam date
- Searches online for summaries and question banks
- **Intensity:** high in the weeks before exams

3. TRIGGERS TO ACT [TR]

- Exam date or test is announced
- New syllabus or lecture PDF received
- Feeling behind classmates
- Recommendation from a friend

10. YOUR SOLUTION [SL]

- **Learnmate – AI StudyBuddy:** upload notes (.txt, .md, .pdf, max 5 MB)
- Google Gemini generates a **summary, flashcards, quiz and study plan**
- Secure accounts with JWT login and student / admin roles
- All results saved in MongoDB so students can revise any time
- Admin panel to manage users and materials
- Automatic retry when the AI service is busy (429/503)

8. CHANNELS OF BEHAVIOR [CH]

- **ONLINE:** web app in a browser on laptop or phone; college WhatsApp / Telegram study groups; social media; GitHub
- **OFFLINE:** word of mouth among classmates; teacher or mentor recommendations; college events and workshops

4. EMOTIONS
BEFORE / AFTER [EM]

- **Before:** overwhelmed, anxious, unprepared
- **After:** confident, organized, motivated